Mechanisms · Tier Pack · Aspire

Learning goal: I can identify levers, pulleys, and gears and explain how they let a small effort move a heavy load.

Aspire Route (Book First)

Compare how a 10-tooth gear driving a 30-tooth gear and a long-armed lever both trade distance/speed for greater force.

Help: The 10-tooth gear must turn __ times to turn the 30-tooth gear once, trading speed for __ turning force.

Optional Make It · Aspire

Make a Gear & Pulley Design Card

Draw a 2-gear train (10 teeth driving 30 teeth) and calculate how many turns the small gear makes for one turn of the large gear.

Word Help

mechanism: a simple machine such as a lever, pulley, or gear that changes a force

lever: a stiff bar that turns on a pivot to lift or move a load

pivot: the fixed resting point (fulcrum) that a lever turns around

pulley: a grooved wheel with a rope that changes the direction or size of a pull

Exit Check

Name the fixed point that a lever turns around.

Where should the pivot be placed to make lifting a heavy load easier?

How do levers, pulleys, and gears allow a smaller force to have a greater effect?